

POC-7 : PV+PVC

STEPS:

1. Create a 1Gi PersistentVolume (hostPath is fine for a local cluster).

```
POC > POC-7-PV-PVC > ! pv.yaml > {} spec > {} hostPath > abc path
1  apiVersion: v1
2  kind: PersistentVolume
3  metadata:
4    name: poc7-pv
5  spec:
6    capacity:
7      storage: 1Gi
8    accessModes:
9      - ReadWriteOnce
10   persistentVolumeReclaimPolicy: Retain
11   hostPath:
12     path: /tmp/poc7-data
```

2. Create a PVC that binds to it.

```
POC > POC-7-PV-PVC > ! pvc.yaml > {} spec > {} resources > {} requests > abc storage
1  apiVersion: v1
2  kind: PersistentVolumeClaim
3  metadata:
4    name: poc7-pvc
5  spec:
6    accessModes:
7      - ReadWriteOnce
8    resources:
9      requests:
10       storage: 1Gi
```

3. Create pod pvc-test (toolbox image), mounting the PVC at /data.

```
POC-7-PV-PVC > ! pod.yaml > {} spec > {} containers > {} 0 > abc image
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: pvc-test
5  spec:
6    containers:
7      - name: toolbox
8        image: docker.io/payal_kharat/kubernetes-toolbox:latest
9        volumeMounts:
10         - name: poc7-storage
11           mountPath: /data
12    volumes:
13      - name: poc7-storage
14        persistentVolumeClaim:
15          claimName: poc7-pvc
```

```

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ ls
pod.yaml pv.yaml pvc.yaml
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl apply -f pv.yaml
persistentvolume/poc7-pv created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pv
NAME      CAPACITY  ACCESS MODES  RECLAIM POLICY  STATUS  CLAIM  STORAGECLASS  VOLUMEATTRIBUTESCLASS  REASON  AGE
poc7-pv   1Gi       RWO           Retain          Available    

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl apply -f pvc.yaml
persistentvolumeclaim/poc7-pvc created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pvc
NAME      STATUS  VOLUME                                     CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
poc7-pvc  Bound   pvc-510e5c8c-669b-4ff0-b17f-3fe398a7be26  1Gi       RWO           standard      <unset>                 3s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl apply -f pod.yaml
pod/pvc-test created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pods
NAME      READY  STATUS   RESTARTS  AGE
pvc-test  1/1    Running  0         5s

```

4. kubectl exec pvc-test -- sh -c 'echo hello-from-poc7 > /data/hello.txt'

```

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl exec pvc-test -- sh -c 'echo hello-from-poc7 > /data/hello.txt'
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl exec pvc-test -- cat /data/hello.txt
hello-from-poc7
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pvc
NAME      STATUS  VOLUME                                     CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
poc7-pvc  Bound   pvc-510e5c8c-669b-4ff0-b17f-3fe398a7be26  1Gi       RWO           standard      <unset>                 3m11s

```

5 . Delete pod pvc-test. Recreate it with the same PVC.

```

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl delete pod pvc-test
pod "pvc-test" deleted from default namespace
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pods
No resources found in default namespace.
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pvc
NAME      STATUS  VOLUME                                     CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
poc7-pvc  Bound   pvc-510e5c8c-669b-4ff0-b17f-3fe398a7be26  1Gi       RWO           standard      <unset>                 10m
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pv
NAME      CAPACITY  ACCESS MODES  RECLAIM POLICY  STATUS  CLAIM  STORAGECLASS  VOLUMEATTRIBUTESCLASS
poc7-pv   1Gi       RWO           Retain          Available    

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl apply -f pod.yaml
pod/pvc-test created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl get pods
NAME      READY  STATUS   RESTARTS  AGE
pvc-test  1/1    Running  0         6s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ kubectl exec pvc-test -- cat /data/hello.txt
hello-from-poc7
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-7-PV-PVC$ |

```